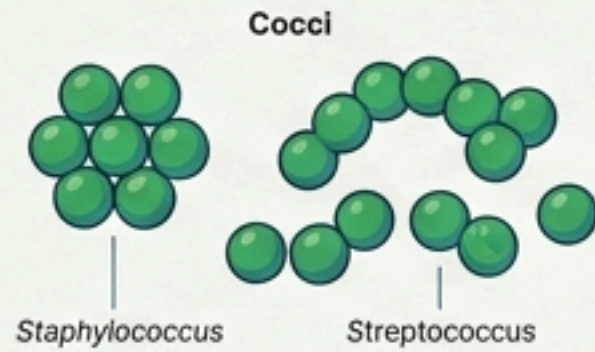
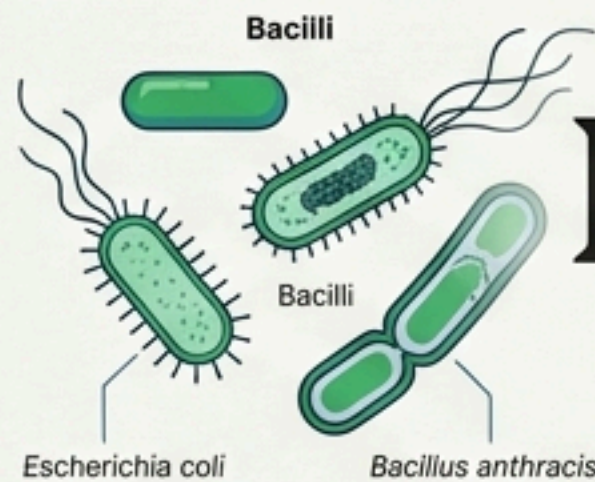


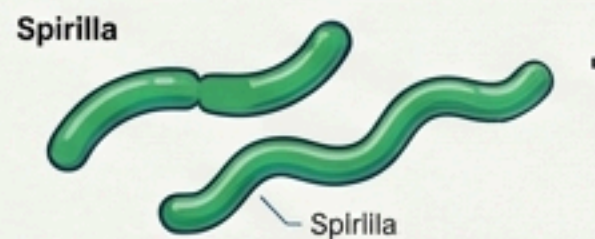
BACTERIAL MORPHOLOGY



- **Spherical bacteria:** are the most common or preferred bacteria.
- **Staphylococcus:** occurs in clusters, such as staphylococci, or common spherical bacteria.
- **Streptococcus:** bacteria: some are flexible staphylococci, common in the throat and in the nose.
- **Byonrasectam:** also called with **neular** bacteria.
- **Escherichia coli** can be a common and **urinary** bacteria.



- **Bacilli:** are bacterial bacteria, such as **oral** cold, including **positive** bacteria.



- **Bacilli:** are bacterial bacteria, such as **oral** cold, including **positive** bacteria.
- **Examples:** *Escherichia coli* bacteria *semelblomray* travellers.

- **Examples:** Rigid: **rigid** rigid characteristically, more such as **positive** bacteria.
- **Examples:** *Treponema pallidum*, *baclnorm* diacteras, like. *Treponema pallidum*.
- **Examples:** Some flexible **on** *spirochetes*, like *helical* and *minor* **positive** bacteria.

Biological Classification: Kingdom Monera

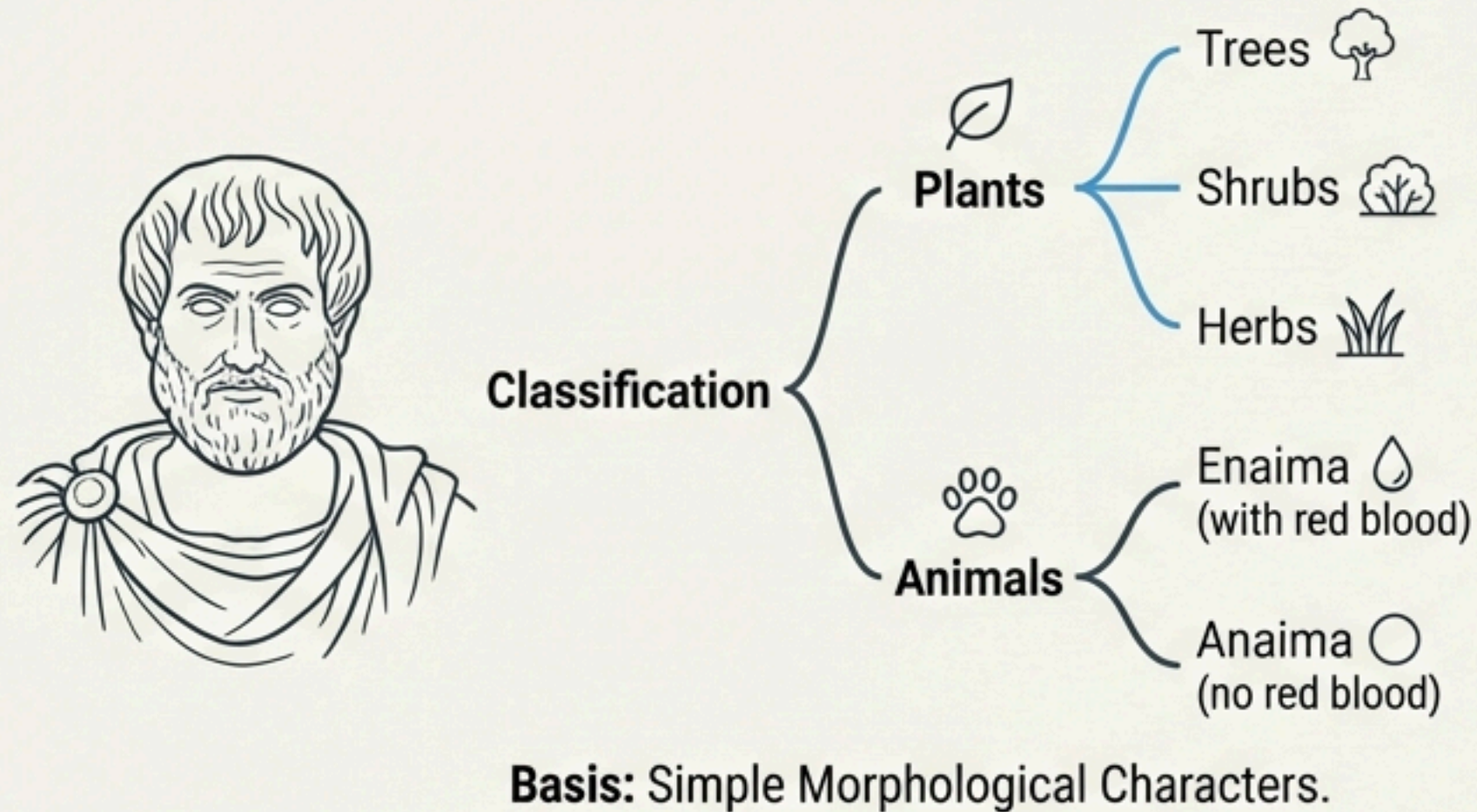
The Textbook-to-Exam Bridge for NEET-UG

Target Audience: Grade 11 & Droppers

Focus: High-Yield NCERT Decoding

Includes 10 Hard-Level Exam Questions for Self-Assessment

The History of Classification: Aristotle to Linnaeus



**Kingdom
Plantae**



**Kingdom
Animalia**



NEET ALERT: Why the Two-Kingdom System Failed

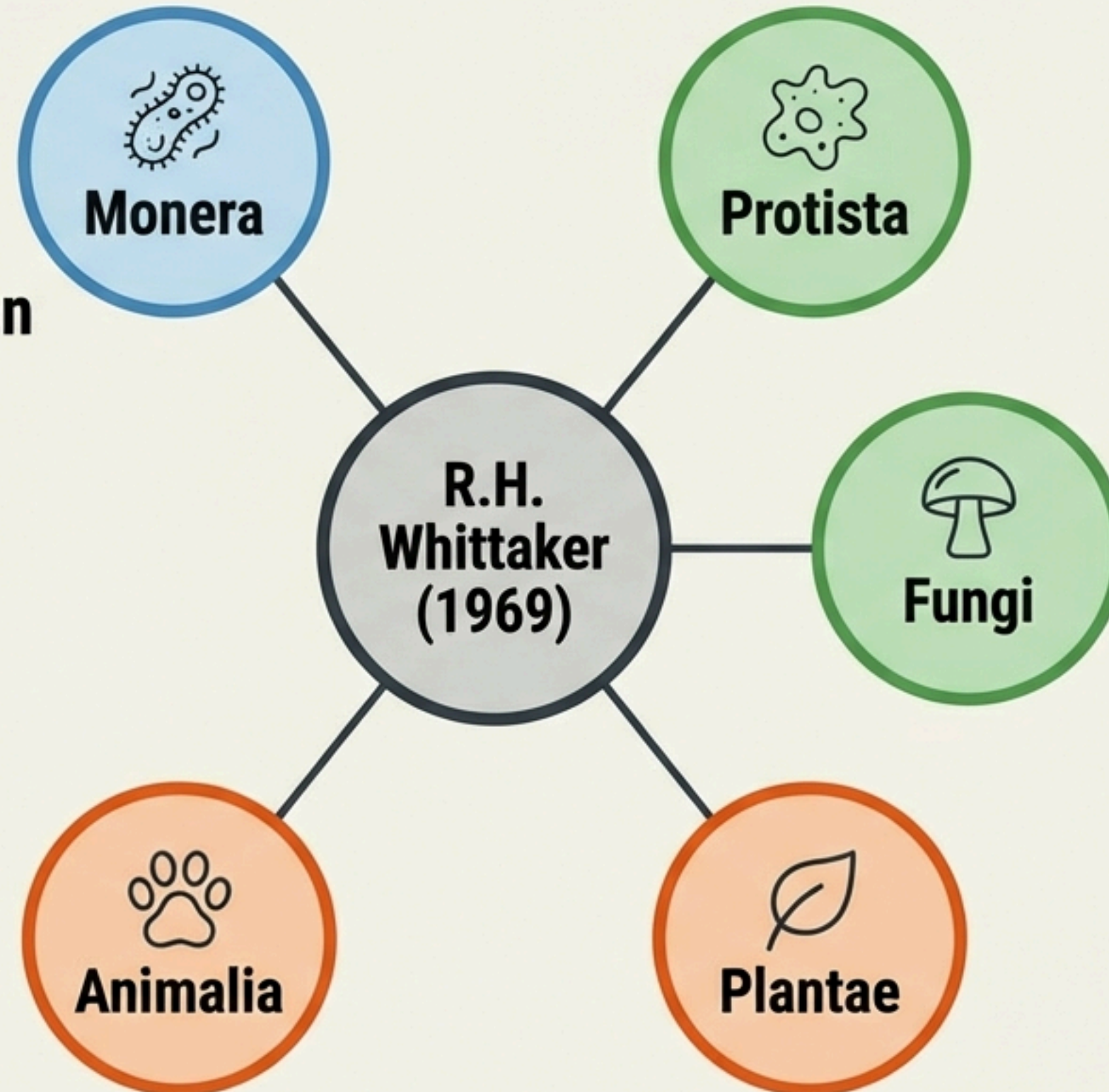
- Did not distinguish between Eukaryotes and Prokaryotes.
- Did not distinguish between Unicellular and Multicellular organisms.
- Did not distinguish between Photosynthetic (e.g., Green Algae) and Non-photosynthetic (e.g., Fungi).

Note: Classification changed because gross morphology was insufficient; cell structure, wall nature, and nutrition became necessary.

R.H. Whittaker's Five Kingdom Classification (1969)

The 5 Main Classification Criteria (Memorize)

- 1. Cell structure
- 2. Body organization
- 3. Mode of nutrition
- 4. Reproduction
- 5. Phylogenetic relationships



Evolution of Logic

Classification changes based on understanding of characteristics and evolutionary relationships.

- Example 1: *Chlamydomonas* & *Spirogyra* were once together (Algae), now separated.
- Example 2: *Amoeba* & *Paramecium* were once "Animals", now Kingdom Protista.

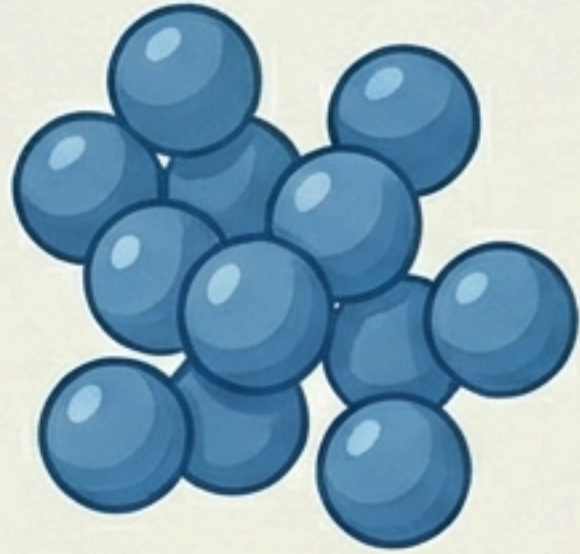
Kingdom Monera vs. The Rest

Comparative Analysis (Based on NCERT Table 2.1)

Characters	Monera	Other Kingdoms (Protista, Fungi, Plantae, Animalia)
Cell Type	Prokaryotic	Eukaryotic
Cell Wall	Non-cellulosic (Polysaccharide + amino acid)	Cellulosic (Plants) or Chitin (Fungi)
Nuclear Membrane	Absent	Present
Body Organization	Cellular	Tissue / Organ / Organ System
Mode of Nutrition	Autotrophic (Chemo/Photo) + Heterotrophic (Sapro/Parasitic)	Specific modes per kingdom

Monera is the ONLY Prokaryotic kingdom with NO nuclear membrane.

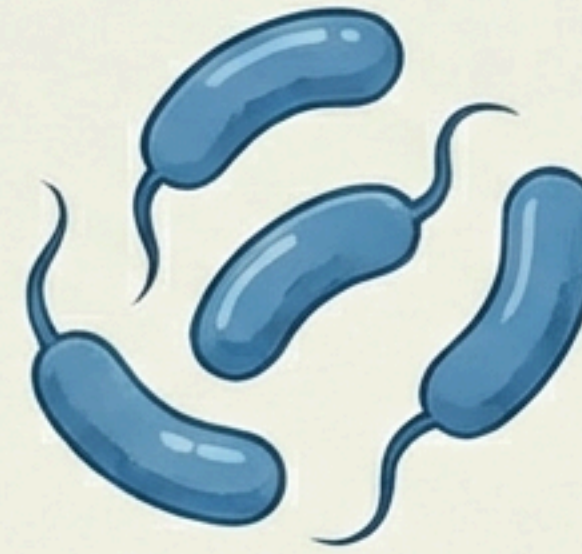
Monera: Sole Members & Shapes



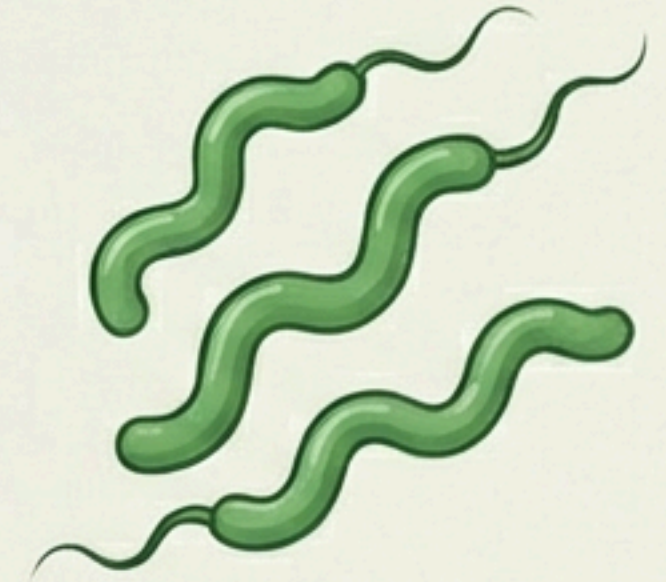
Coccus



Bacillus



Vibrio



Spirillum

Habitat: Ubiquitous

Found in soil, hot springs, deserts, deep oceans, and as parasites.

NEET ALERT - High Yield Statement

“Though the bacterial structure is very simple, they are very complex in behaviour.”

Bacteria as a group show the **MOST EXTENSIVE** metabolic diversity (Autotrophic + Heterotrophic).

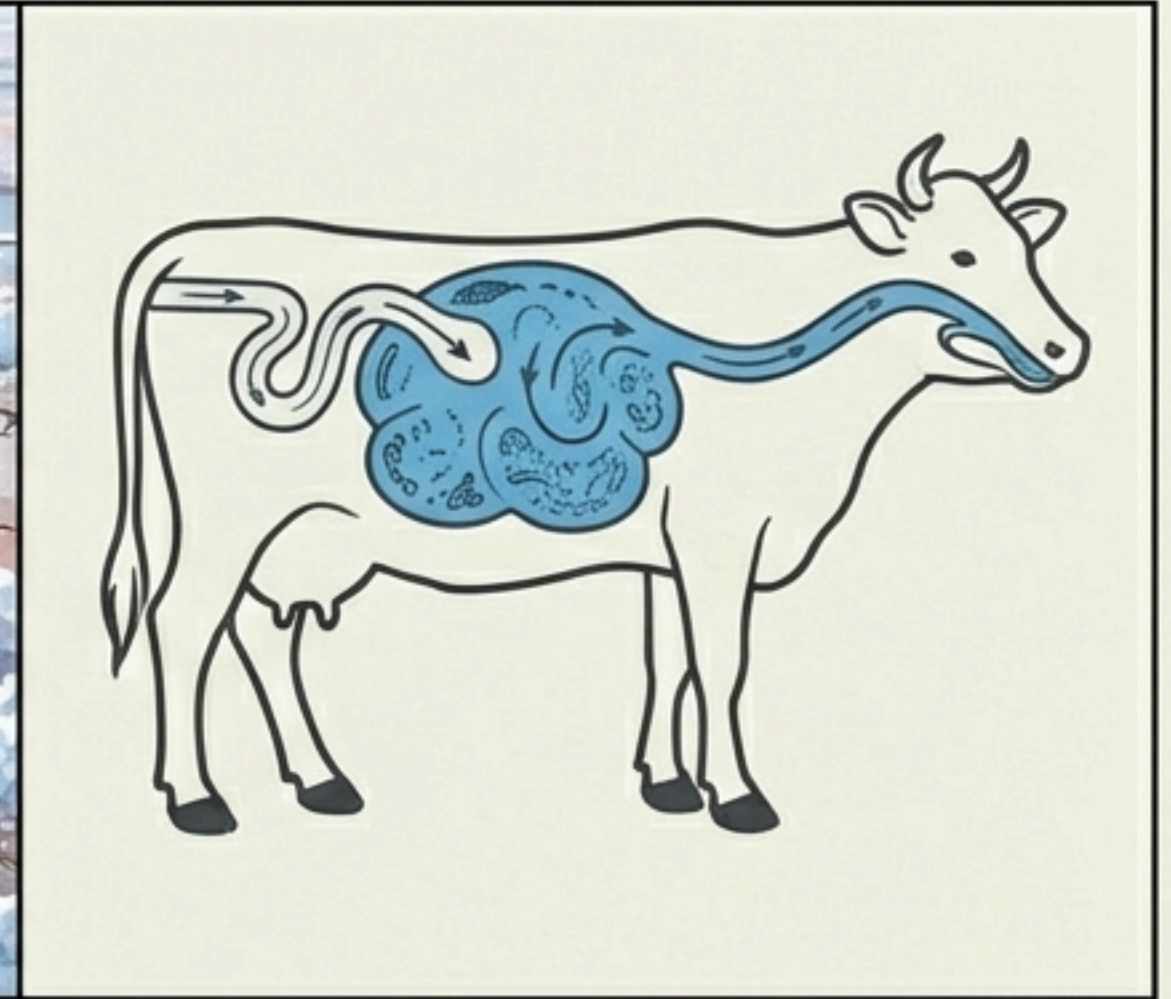
Archaeobacteria: The Extremophiles



Thermoacidophiles
(Hot Springs)



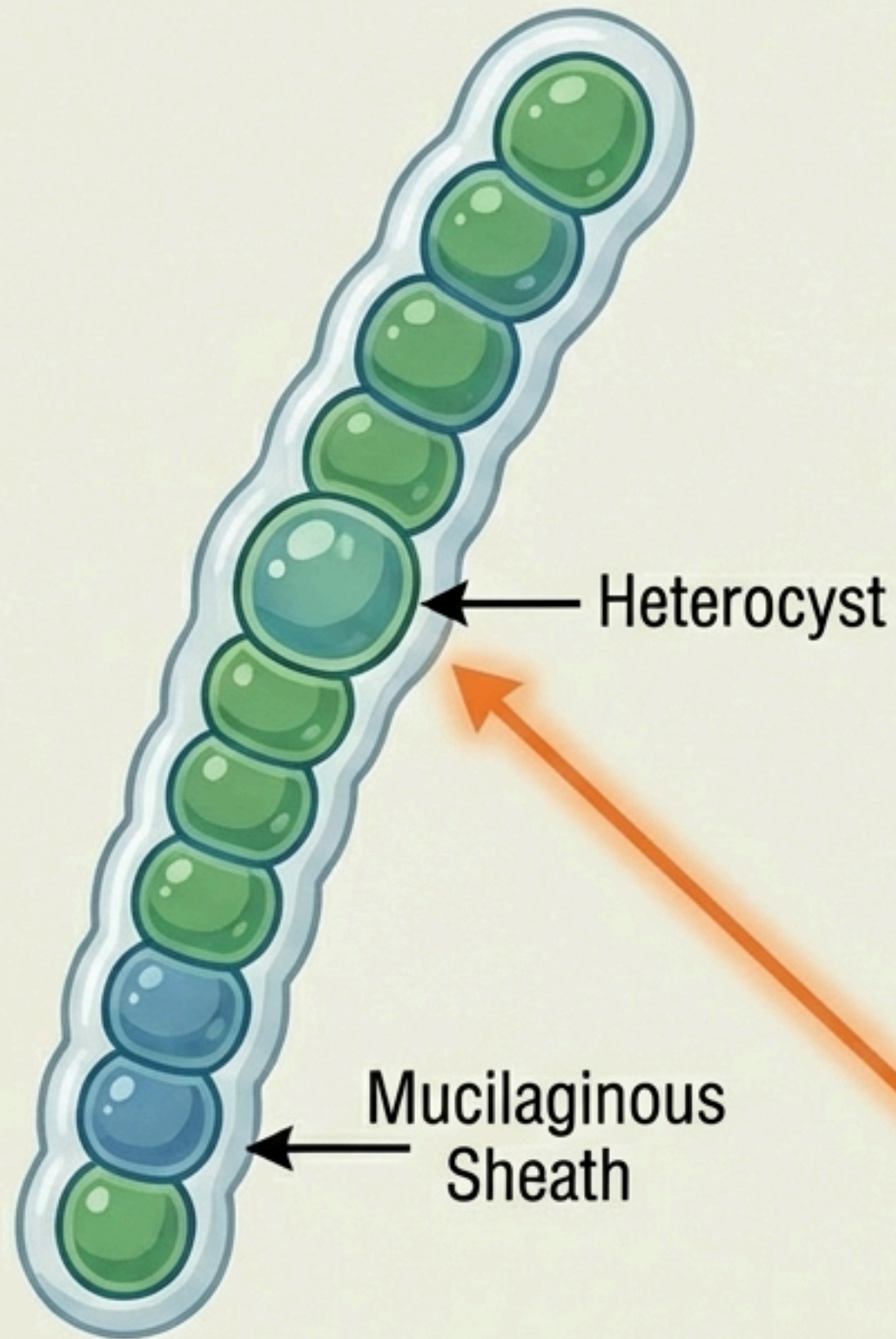
Halophiles
(Salty Areas)



Methanogens
(Marshy areas & Ruminant Gut)

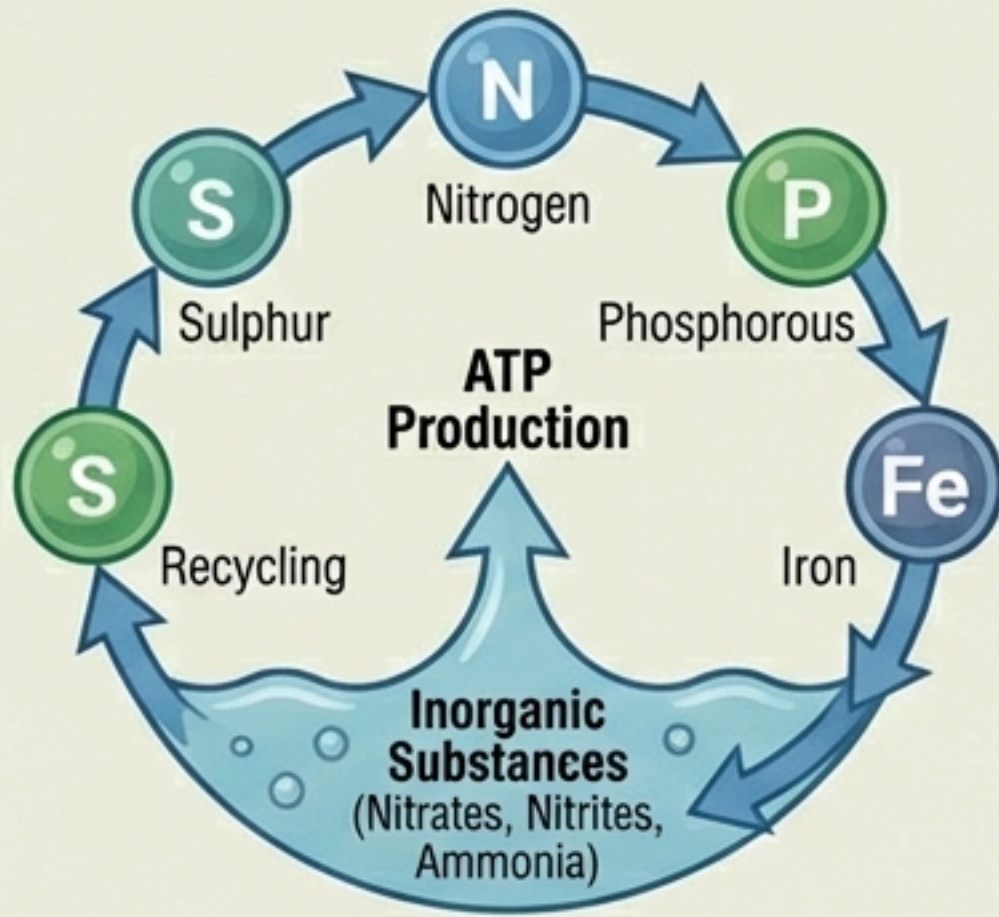
- **Defining Feature:** Different cell wall structure compared to other bacteria. This feature is responsible for their survival in extreme conditions.
- **Application:** Methanogens in the gut of cows/buffaloes produce methane (biogas) from dung.

Eubacteria: Cyanobacteria (Blue-Green Algae)



- **Photosynthetic Autotrophs:** Have chlorophyll 'a' similar to green plants.
- **Forms:** Unicellular, colonial, or filamentous.
- **Ecological Impact:** Often form 'blooms' in polluted water bodies.
- **Nitrogen Fixation:** Fix atmospheric nitrogen in specialized cells called **Heterocysts** (e.g., Nostoc, Anabaena).

Eubacteria: The Recyclers and The Decomposers



Chemosynthetic Autotrophs

- Oxidize inorganic substances (Nitrates, Nitrites, Ammonia) to produce ATP.
- **Role:** Great importance in recycling nutrients (Nitrogen, Phosphorous, Iron, Sulphur).

Heterotrophic Bacteria

Most abundant in nature. The majority are important decomposers.

Helpful (Beneficial)

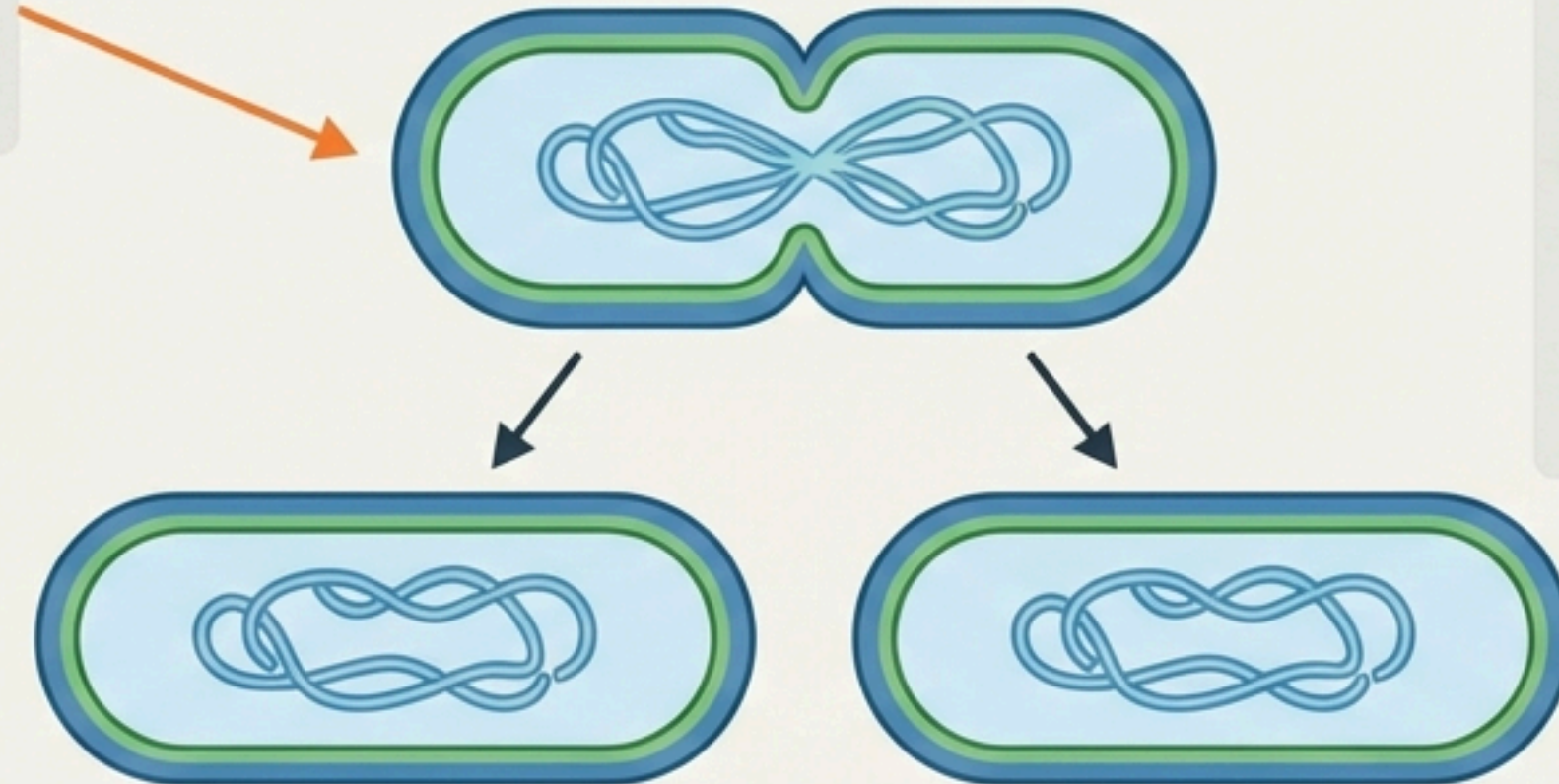
- Making curd (Lactobacillus)
- Antibiotics
- N-fixation in legumes.

Harmful/Pathogenic (Disease-Causing)

- **Cholera**
- **Typhoid**
- **Tetanus**
- **Citrus Canker.**

Bacterial Reproduction

Main Method:
Fission



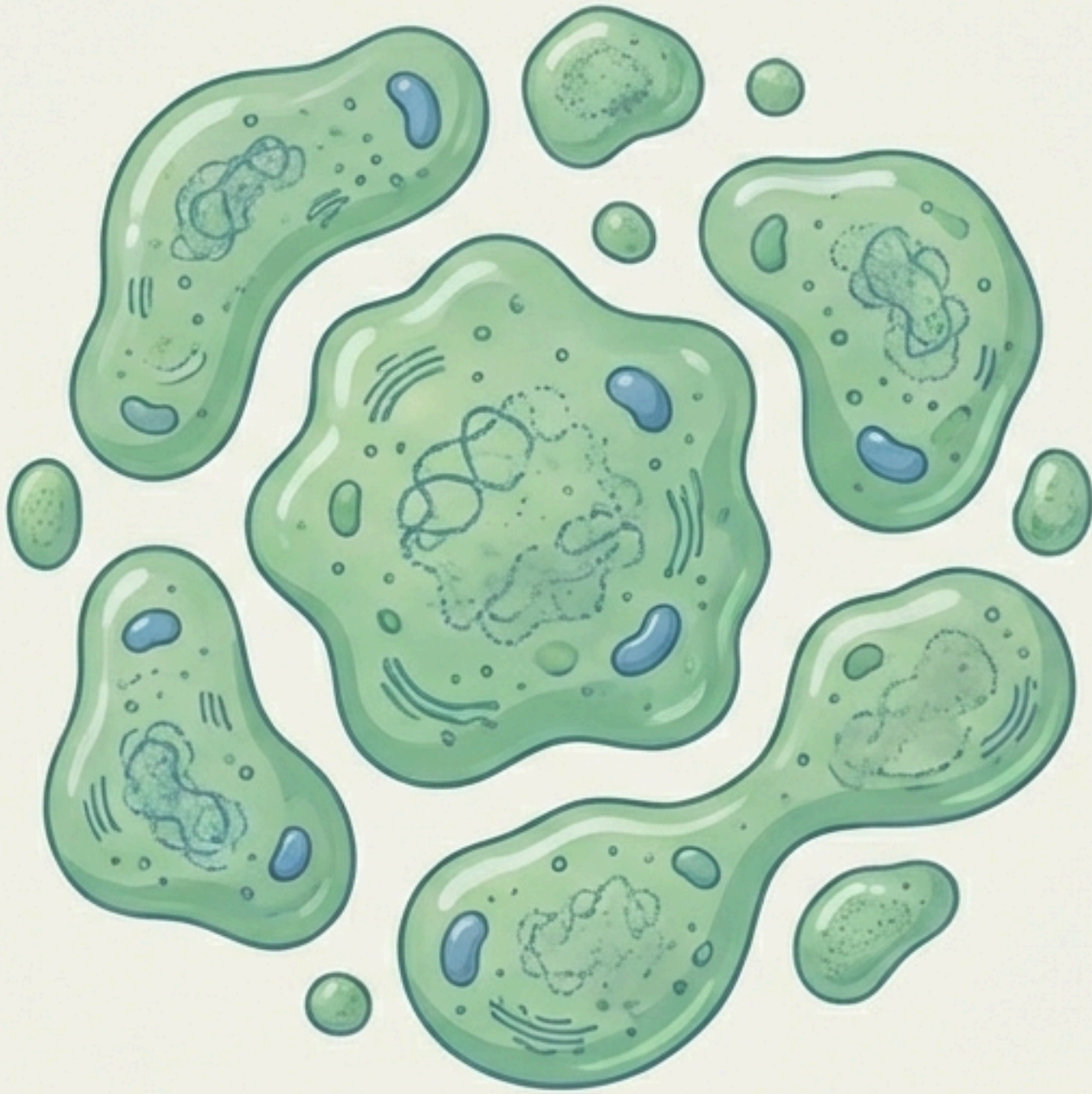
Unfavorable
Conditions:
Spore Formation



Sexual Reproduction

Not gamete fusion. It is a primitive type of DNA transfer from one bacterium to another.

The Anomalies: Mycoplasma



- **Completely LACK a cell wall.**
- Smallest living cells known.
- Can survive **WITHOUT** oxygen (Anaerobic).
- Many are pathogenic in animals and plants.

NEET ALERT

Common 'Odd One Out'
Question Target.

NEET Challenge: Section A (Statement Based)

Q1

Read the following statements: (a) Bacteria are sole members of Monera. (b) Bacteria show least metabolic diversity. (c) Bacterial structure is complex but behavior is simple. Which are correct?

Answer: Only (a). [Reason: Most metabolic diversity; Simple structure/Complex behavior]

Q2

Identify the INCORRECT statement regarding Cyanobacteria: A) Have chlorophyll "a". B) Form blooms in polluted water. C) Fix nitrogen in all vegetative cells. D) Surrounded by gelatinous sheath.

Answer: C. [Reason: Fixation occurs only in Heterocysts]

Q3

Which organism completely lacks a cell wall and is anaerobic? A) Nostoc. B) Mycoplasma. C) Vibrio. D) Yeast.

Answer: B.

NEET ALERT



Common "Odd One Out" Question Target.

NEET Challenge: Section B (Assertion & Reason)

(A) Both True & Correct Exp. (B) Both True & Incorrect Exp. (C) A True, R False. (D) Both False.

Q4

Assertion: Archaeobacteria survive in harsh habitats like hot springs.

Reason: They possess a different cell wall structure compared to other bacteria.

Answer: A.

Q5

Assertion: The Two Kingdom classification was found inadequate.

Reason: It failed to distinguish between eukaryotes/prokaryotes and uni/multicellular organisms.

Answer: A.

Q6

Assertion: Chemosynthetic bacteria use light energy to synthesize food.

Reason: They play a great role in recycling N, P, Fe, and S.

Answer: D (technically Assertion is False).

NEET Challenge: Section C (Critical Application)

Q7

Q7: Match the organism to its characteristic:

- | | |
|------------------------|------------------|
| (i) Halophiles | (a) Salty Areas |
| (ii) Methanogens | (b) Ruminant Gut |
| (iii) Cyanobacteria | (c) Heterocysts |
| (iv) Thermoacidophiles | (d) Hot Springs |

Answer: i-a, ii-b, iii-c, iv-d

Q8

Q8: Which set consists ONLY of bacterial diseases?

- A) Cholera, Typhoid, Tetanus, Citrus Canker
B) Mumps, Cholera, Typhoid
C) Tetanus, Citrus Canker, Smallpox

Answer: A.

Q9

Q9: In Whittaker's system, which feature is ABSENT in Monera but present in all others?

- A) Cell Wall. B) Nuclear Membrane.
C) Ribosomes.

Answer: B.

Final Challenge & Summary

The Final Trap Question

Q10: Under unfavorable conditions, bacteria reproduce primarily by:
A) Fission. B) Spores. C) Gametes. D) DNA Transfer.

Answer: B. [Note: Fission is favorable; Spores are unfavorable].

High-Yield Recap

- **Monera:** Prokaryotic, Unicellular, No Nucleus.
- **Archaeobacteria:** Extreme walls (Salt/Heat/Methane).
- **Eubacteria:** Cyanobacteria (N-fix/Heterocyst) + Heterotrophs (Decomposers).
- **Mycoplasma:** No wall, Anaerobic, Smallest.

Master the Text, Master the Exam.

Every line in this presentation is a potential question. Review the 'NEET ALERT' slides before your exam.



[] Kingdom Monera: Complete.